Schedule

- PS this Thursday:
  - Case exam 1
  - Study/practice questions
- LearnHub: Class exercises, reading assignments and practice problems

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Agenda

- Regression review
- Estimating and predicting using the regression line
- Significance of LR
- Variability explained by LR
- Correlation

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Simple linear regression (1)

- The sample: $\{(x_i, y_i)\}_{i=1}^{n}$

independent variable

dependent variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

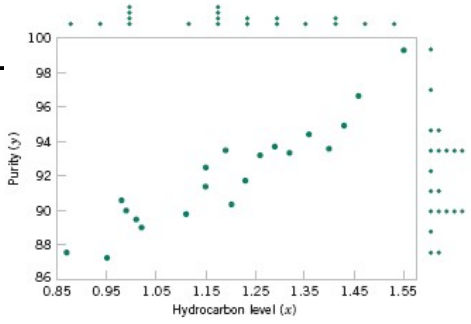
$$\varepsilon \sim N(0, \sigma^2)$$

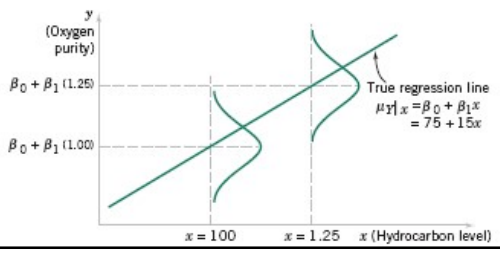
random deviation or random error

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Empirical Models

Observation Number	Hydrocarbon Level x(%)	Purity y(%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33





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$y_i = \beta_0 + \beta_1 x_i + \varepsilon$

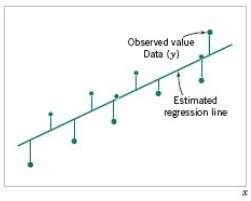
What are we interested in?

Building the model

- Estimating β_0 and β_1

→ the fitted regression line

b_0, b_1: Minimize the sum of squared deviations.



- Estimating σ
- Distribution of b_0, b_1, $\hat{Y} = \hat{\mu}_{Y|x_0} = b_0 + b_1 x_0$

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Distribution of b_0 ?

$$b_0 = \overline{y} - b_1 \overline{x}$$

$$E(b_0) = \beta_0$$

$$V(b_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

$$b_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)$$

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Distribution of b_1 ?

$$b_1 = \frac{S_{xy}}{S_{xx}}$$

$$E(b_1) = \beta_1$$

$$V(b_1) = \frac{\sigma^2}{S_{xx}}$$

$$b_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

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Estimating σ^2 (1)

$$\varepsilon_i \sim N(0, \sigma^2)$$

variability of each of Y_i about the true regression line.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

variability of data pts about the fitted regression line.

$$\hat{\sigma}^2 = ?$$

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Estimating σ^2 (2)

$two \quad parameters \quad to \quad estimate \quad \beta_0, \beta_1$

$$E[SSE] = (n-2)\sigma^2 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{SSE}{n-2} = MSE$$

can be shown

Two parameters to Estimate β_0, β_1 $\Rightarrow$ a loss of 2 df

MSE is an unbiased estimator for σ^2.

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Distribution of b_1 ?

$$b_1 = \frac{S_{xy}}{S_{xx}}$$

$$E(b_1) = \beta_1$$

$$V(b_1) = \frac{\sigma^2}{S_{xx}}$$

$$b_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$T = \frac{b_1 - \beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} = \frac{b_1 - \beta_1}{S_{b_1}} \sim t_{n-2}$$

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$y_i = \beta_0 + \beta_1 x_i + \varepsilon$

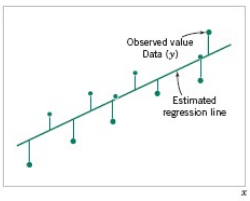
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- Estimating σ
- Distribution of b_0, b_1, $\hat{Y} = \hat{\mu}_{Y|x_0} = b_0 + b_1 x_0$

Evaluating the model

- Is the model significant?
- How useful is the model?

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Is the linear regression model significant?

$H_0: \beta_1 = 0$ $\quad$ $H_a: \beta_1 \neq 0$

Test statistics:

$$T = \frac{b_1}{s_{b_1}} \sim t_{n-2}$$

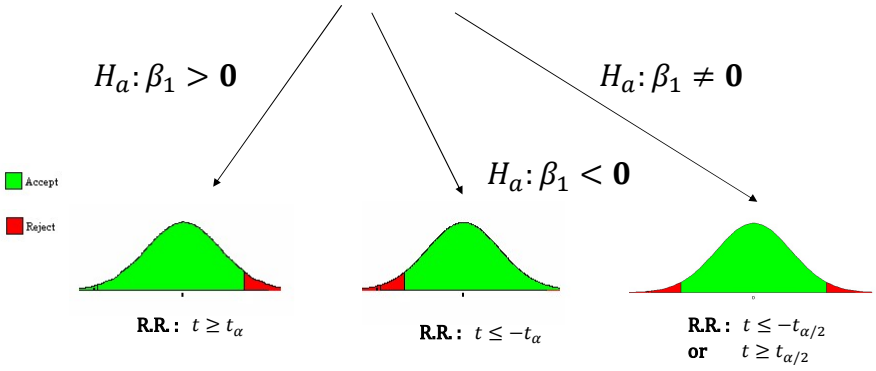


R.R.: $t \leq -t_{\alpha/2}$
or $t \geq t_{\alpha/2}$

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Hypothesis testing on β_1

$H_0: \beta_1 = \mathbf{0}$ Test statistics: $T = \dfrac{b_1}{s_{b_1}} \sim t_{n-2}$



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ANalysis Of Variance (ANOVA) table for simple regression

Source	*df*	*Sum Sq.*	*Mean Sum Sq.*	*F*
Regression	*1*	*SSR*	$MSR = \frac{SSR}{1} = SSR$	$\frac{SSR}{MSE}$
Error	*n-2*	*SSE*	$MSE = \hat{\sigma}^2 = \frac{SSE}{n-2}$	
Total	*n-1*	*SST*		

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ANalysis Of VAriance identity

$$SST = S_{yy} = \sum (y_i - \overline{y})^2 = \sum y_i^2 - \frac{\sum y_i^2}{n}$$ total variability of Y.

$$SSE = \sum (y_i - \hat{y}_i)^2$$ error sum of squares
Interpretation: unexpected variability of Y in regression.

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$ regression sum of squares
Interpretation: amount of variability that is explained by the model.

$$SST = \sum (y_i - \overline{y})^2 = \sum (\hat{y}_i - \overline{y})^2 + \sum (y_i - \hat{y}_i)^2$$

df=n-1 df=1 df=n-2

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ANalysis Of VAriance identity

$E(SSE/(n-2)) = \sigma^2$ $SSE/\sigma^2 \sim \chi^2_{n-2}$

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ $SSR/\sigma^2 \sim \chi^2_1$

If β_1=0, what is the distribution of $\frac{SSR/1}{SSE/(n-2)}$?

$\frac{W/u}{Y/v} \sim F_{u,v}$ $W \sim \chi^2_u$, $Y \sim \chi^2_v$

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ANalysis Of Variance (ANOVA) table for simple regression

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Regression	*1*	*SSR*	$MSR = \frac{SSR}{1} = SSR$	$\frac{SSR}{MSE}$
Error	*n-2*	*SSE*	$MSE = \hat{\sigma}^2 = \frac{SSE}{n-2}$	
Total	*n-1*	*SST*		

$H_0: \beta_1 = 0$

$H_a: \beta_1 \neq 0$

→ can be tested via:

$t^2 = f \quad \Rightarrow \quad t^2_{\alpha/2,n-2} = F_{\alpha,1,n-2}$

reject H_0 if $F > F_{\alpha,1,n-2}$.

↓

equivalent to t test in simple linear regression. F-test is only two-sided.

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Is the linear regression model significant?

$H_0: \beta_1 = 0 \qquad H_a: \beta_1 \neq 0$

Test statistics:

$T = \frac{b_1}{s_{b_1}} \sim t_{n-2}$



Accept Reject

R.R.: $t \leq -t_{\alpha/2}$ **or** $t \geq t_{\alpha/2}$

$t^2 = f \quad \Rightarrow \quad t^2_{\alpha/2,n-2} = F_{\alpha,1,n-2}$

reject H_0 if $F > F_{\alpha,1,n-2}$.

only two-sided

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Inferences concerning $\mu_{Y|x0}$ (1)

$$\hat{Y} = \hat{\mu}_{Y|x_0} = b_0 + b_1 x_0 \quad \text{~normal}$$

$$E(\hat{Y}) = \beta_0 + \beta_1 x_0$$

$$V(\hat{Y}) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$$\hat{V}(\hat{Y}) = \hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

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Theorem

The variable:

$$T = \frac{b_0 + b_1 x* - (\beta_0 + \beta_1 x*)}{S_{b_0 + b_1 x*}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x*)}{S_{\hat{Y}}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x*)}{\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}$$

has a t−distribution with degrees of freedom n−2.

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Prediction interval for a future value of Y

- Error of prediction: $Y - (b_0 + b_1 x_0)$

$$E(Y - (b_0 + b_1 x_0)) = 0$$

$$V(Y - (b_0 + b_1 x_0)) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)$$

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Theorem

The variable:

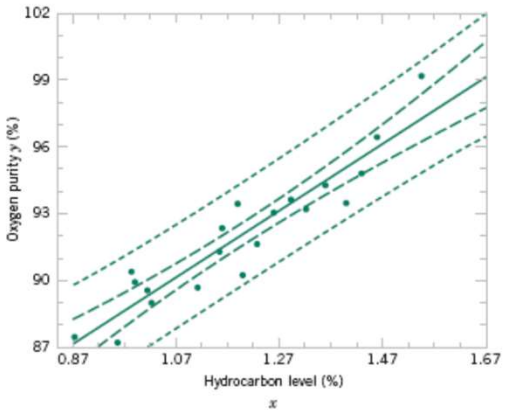
$$T = \frac{Y - (b_0 + b_1 x_0)}{S\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} = \frac{Y - (b_0 + b_1 x_0)}{\sqrt{S^2 + S^2_{b_0 + b_1 x_0}}} = \frac{Y - (b_0 + b_1 x_0)}{\sqrt{S^2 + S^2_{\hat{Y}}}},$$

with $S = \hat{\sigma}$, has a t−distribution with degrees of freedom n−2.

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Example

Scatter diagram of oxygen purity data with fitted regression line, 95% prediction limits (outer lines) , and 95% confidence limits (inner lines) on $\mu_{Y|x0}$.



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Important issues

- Is the linear regression model significant?
- How much of the variability in Y (dependent variable) is explained by X (independent variable)?

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How much of the variability in Y is explained by X?

- The sample: $\{(x_i, y_i)\}_{i=1}^n$
- LR model: $Y = \underbrace{\beta_0 + \beta_1 X}_{\text{True regression line}} + \varepsilon$ (Random errors) $\quad \varepsilon \sim N(0, \sigma^2)$

$$y_i = \underbrace{b_0 + b_1 x_i}_{\text{Estimated regression line}} + e_i \quad (e_i: \text{residuals}) \qquad SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - (b_0 + b_1 x_i)\right)^2$$

$$SST = \sum (y_i - \overline{y})^2 = \underbrace{\sum (\hat{y}_i - \overline{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

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How much of the variability in Y is explained by X?

- The sample: $\{(x_i, y_i)\}_{i=1}^n$
- LR model: $\varepsilon \sim N(0, \sigma^2)$

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Proportion of observed y variation that can be explained by the simple linear regression model.

measures the degree of linear relationship between X and Y

$$SST = \sum (y_i - \overline{y})^2 = \underbrace{\sum (\hat{y}_i - \overline{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

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Coefficient of determination

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Proportion of observed y variation that can be explained by the simple linear regression model.

measures the degree of linear relationship between X and Y

r^2 has no relation with the magnitude of the slope of regression line.

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Example : Computer software

Regression Analysis

The regression equation is

Purity = 74.3 + 14.9 HC Level

Predictor	Coef	SE Coef	T	P
Constant	74.283 ← $\hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 ← $\hat{\beta}_1$	1.317	11.35	0.000

S = 1.087 R-Sq = 87.7% R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 ← SS_E	1.18 ← $\hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

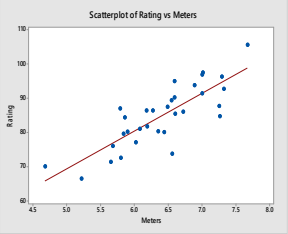
Values of Predictors for New Observations

New Obs	HC Level
1	1.00

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Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).



Regression Analysis: Rating versus Meters

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	1669.02	1669.02	61.02	0.000
Error	30	820.50	27.35		
Total	31	2489.52			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
5.22972	67.04%	65.94%	62.11%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	14.19	9.09	1.56	0.129
Meters	11.04	1.41	7.81	0.000

Regression Equation

Rating = 14.19 + 11.04 Meters

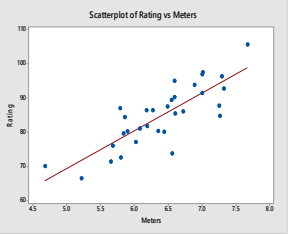
Is this regression model significant?

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Class Exercise

$$T = \frac{b_1-\beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} = \frac{b_1-\beta_1}{S_{b_1}} \sim t_{n-2}$$

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Can you find a 95% confidence interval for β1?

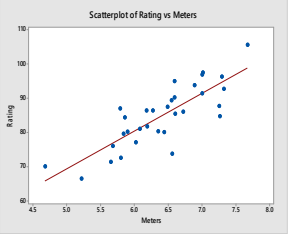
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$b_0 = \overline{y} - b_1\overline{x}$

Class Exercise

$$T = \frac{\hat{Y}-(\beta_0+\beta_1 x*)}{S_{\hat{Y}}} = \frac{\hat{Y}-(\beta_0+\beta_1 x*)}{\hat{\sigma}\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}} \sim t_{n-2}$$

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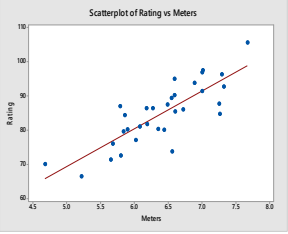
Can you find a 95% confidence interval around Rating for Meters=7?

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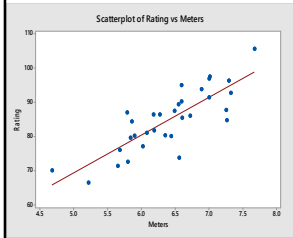
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Regression Equation

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How much of the variability in Rating is explained by Meters?

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Correlation (1)

Population

$$Cov(X,Y) = E((X-\mu_X)(Y-\mu_Y))$$

$$Cov(X,Y) = E(XY) - \mu_X\mu_Y$$

$$Corr(X,Y) = \rho_{XY} = \frac{Cov(X,Y)}{\sigma_X\sigma_Y}$$

population correlation coefficient

Sample

$$S_{xy} = \sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$

$$\frac{S_{xy}}{n} = \frac{\sum x_i y_i}{n} - \bar{x}\bar{y}$$

$$r_{xy} = r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$$

sample correlation coefficient

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Correlation (2)

Properties	Properties
$Corr(aX+b, cY+d) = Corr(X,Y) = Corr(Y,X)$	$r_{ax+b,cy+d} = r_{x,y} = r_{y,x}$
$-1 \le \rho_{XY} \le 1$	$-1 \le r_{xy} \le 1$
$\rho_{xy} = 1$ or $\rho_{xy} = -1$ if $Y = aX + b$.	$r_{xy} = 1$ or $r_{xy} = -1$ if $y_i = ax_i + b$.
$\rho_{xy} = 0$ if X, Y are independent.	$(r_{xy})^2 = (r)^2 = r^2$ coefficient of determination

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Interpretation of r

- r^2 is not related with the magnitude of the slope.
- When r^2 is small, an alternative model should be seeked: Generally, one looks for other independent variables which can explain a bit more of the variation in Y.

  → multiple regression

- Interpretation of value of r^2 changes with the context:
  - physical/engineering: 80%
  - social/psychology: 50% can be good enough

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